MULTI-DEPARTMENT OFFICE NETWORK WITH RIP ROUTING AND SECURE REMOTE ACCESS

1. Introduction

This project focuses on designing and configuring a multi-department office network using Cisco Packet Tracer. The network consists of seven departments: HR, Finance and Accounting, Marketing, Sales, Operations, IT, and Customer Service.

Each department has its own router, switch, server, and connected computers. The network is designed to allow communication between computers within the same department and between different departments.

The RIP (Routing Information Protocol) is used to exchange routing information between routers. DHCP is used to assign IP addresses to computers automatically. Telnet and SSH are used to demonstrate remote access to network devices and servers, depending on the configuration.

2. Objectives

- To design a network for seven departments of an office.
- To connect each department using routers and switches.
- To provide a separate server for each department.
- To configure DHCP for automatic IP address assignment.
- To implement RIP for communication between different departments.
- To configure Telnet and SSH for remote access.
- To understand network connectivity, routing, and basic network administration.

3. Software and Devices Used

Software: Cisco Packet Tracer

Network devices:

- 7 routers
- 7 switches
- 7 servers
- 14 PCs
- Appropriate Ethernet and serial connections

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4. Network Topology

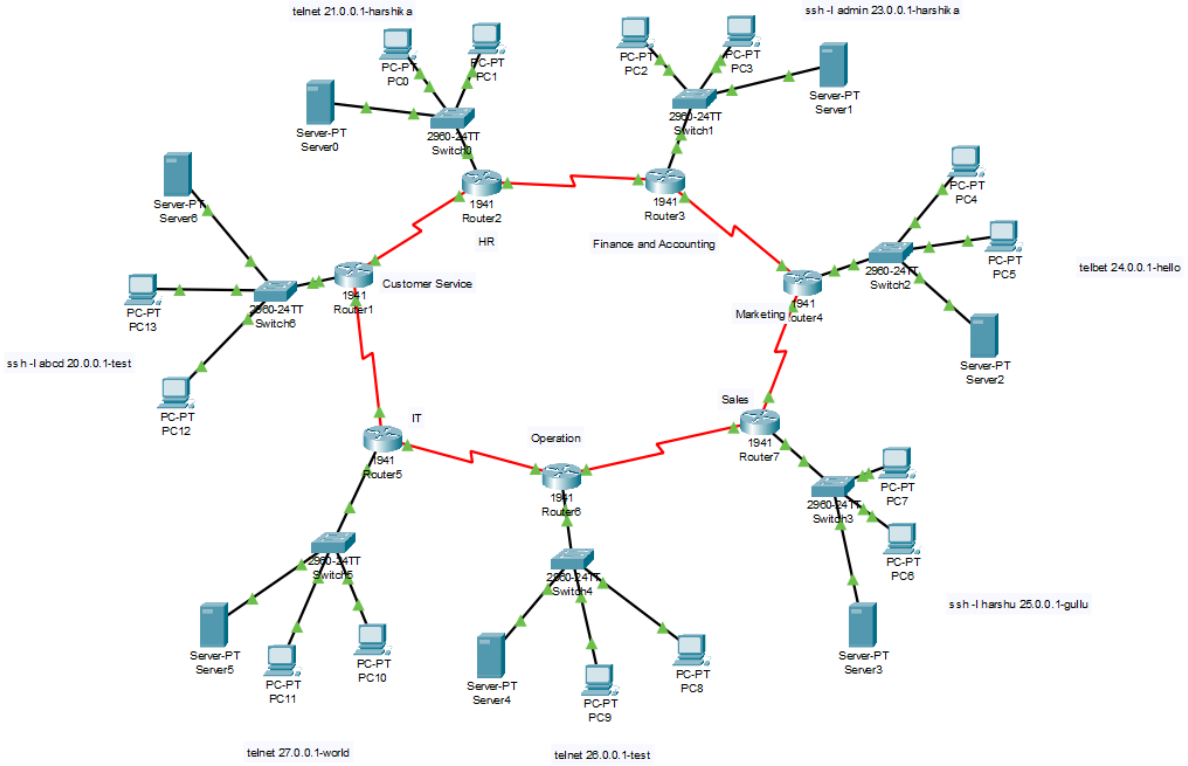
The project uses a ring-shaped arrangement of seven routers. Each router represents one department and connects to its departmental switch. The switches connect the department's PCs and server.

The routers are connected in a loop, allowing routing information to be shared throughout the network. This arrangement provides multiple router-to-router links for communication between departments.

Departments included:

1. HR
2. Finance and Accounting
3. Marketing
4. Sales
5. Operations
6. IT
7. Customer Service

Figure 1: Cisco Packet Tracer topology of the seven-department office network.



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5. Working of the Network

5.1 Departmental Network

Each department has a dedicated router and switch. The PCs and server are connected to the departmental switch. The router connects the departmental LAN to the other departmental networks.

5.2 Server Configuration

A separate server is provided for each department. The server can provide services to the connected departmental computers, depending on its configuration. Each server is assigned an IP address so that devices can communicate with it over the network.

5.3 DHCP Configuration

DHCP (Dynamic Host Configuration Protocol) automatically assigns IP configuration to client computers. This reduces the need to configure every PC manually.

The DHCP configuration can provide:

- IP address
- Subnet mask
- Default gateway
- DNS server address, if configured

5.4 RIP Routing

RIP (Routing Information Protocol) is used to exchange routing information between the seven routers. It allows routers to learn routes to networks connected to other departments.

When a PC sends data to a PC or server in another department, the packet travels through the appropriate router path based on the routing table. RIP uses hop count as its routing metric.

5.5 Telnet Remote Access

Telnet provides remote command-line access to a network device over a TCP connection. It can be used to demonstrate remote administration. However, Telnet transmits data, including login credentials, without encryption, so it is not recommended for secure production networks.

5.6 SSH Remote Access

SSH (Secure Shell) provides encrypted remote command-line access to supported network devices. It protects login credentials and session data while they travel across the network.

SSH is preferred over Telnet when secure remote administration is required.

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6. Testing and Verification

The following tests can be performed in Cisco Packet Tracer to verify the network:

1. **Local connectivity:** Ping between two PCs in the same department.
2. **Interdepartmental connectivity:** Ping a PC in one department from a PC in another department.
3. **Server connectivity:** Ping the departmental server from a connected PC.
4. **DHCP verification:** Check whether PCs receive valid IP addresses automatically.
5. **RIP verification:** Check the routing tables and confirm that remote departmental networks are learned.
6. **Telnet verification:** Connect to a device using its configured Telnet access.
7. **SSH verification:** Connect to a device using its configured SSH access.

Successful ping replies and remote login sessions indicate that the relevant configurations are working correctly.

7. Advantages

- Enables communication between seven departments.
- Provides separate departmental networks and servers.
- Reduces manual IP configuration through DHCP.
- Allows routers to learn routes dynamically through RIP.
- Demonstrates both basic and encrypted remote administration.
- Provides practical experience in network design and troubleshooting.

8. Limitations

- RIP has a maximum usable hop count of 15, which limits its suitability for larger networks.
- Telnet is not secure because it does not encrypt communication.
- Separate departmental servers require correct IP addressing and service configuration.
- Network communication depends on correct router, switch, and routing configurations.
- A ring-shaped physical or logical layout does not automatically guarantee uninterrupted connectivity if a link or router fails.

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9. Conclusion

This project demonstrates the design and configuration of a multi-department office network using Cisco Packet Tracer. It integrates seven departmental networks, dedicated servers, DHCP, RIP routing, and remote access through Telnet and SSH.

The project helps develop practical knowledge of IP addressing, network connectivity, dynamic routing, server communication, and remote administration. It also highlights the importance of using secure protocols such as SSH for managing network devices.

10. Future Scope

The network can be improved by implementing VLANs, access control lists (ACLs), firewall rules, and stronger security policies. RIP could also be replaced with OSPF to explore a more scalable routing protocol. Additional monitoring with Wireshark could be used to study packet flow, routing updates, and Telnet-versus-SSH traffic.

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